

Object Oriented Programming- Fall-2023

Assignment 1

Pointers and Dynamic Memory

Due Date : 16th September 2023

Total Marks : 100

Instructions

- No Marks for code with errors in any questions
- Submit single **Zip file**, having all of your questions in cpp files.
- Naming convention should be followed for zip file as **22F_XXXX_A1_Sec.zip**
- No extensions in deadline.

1- 1D Pointers: (10*2 = 20 Marks)

1. **Number Conversion:** Write a C++ program that converts a given decimal number into binary, octal, and hexadecimal representations using 1D pointers for character storage. Ensure proper memory allocation and deallocation.

Sample input

Enter a decimal number: 63

Sample output

Binary: 111111

Octal: 77

Hexadecimal: 3F

2. **String Reversal:** Develop a program that reverses a given string using 1D pointers. Implement a function that takes the input string and returns the reversed string without using any string library functions.

II- 2D Pointers: (15*2 = 30 Marks)

3. **Matrix Operations:** Create a C++ program that performs matrix addition and multiplication using 2D pointers. Users should input two matrices, and the program should dynamically allocate memory to store the matrices, perform the chosen operation, and display the result.

Sample Input

Enter the number of rows and columns for Matrix A: 2 2

Enter elements for Matrix A:

1 2

3 4

Enter the number of rows and columns for Matrix B: 2 2

Enter elements for Matrix B:

5 6

7 8

Enter 1 for addition or 2 for multiplication: 1

Sample Output

Matrix Addition Result:

6 8

10 12

4. **2D Array Manipulation:** Implement a program that reads a 2D array of integers from the user and performs operations such as finding the sum, average, and standard deviation of the elements in each row and column using 2D pointers.

Sample Input

Enter the number of rows and columns for the 2D array: 3 3

Enter elements for the 2D array:

1 2 3

4 5 6

7 8 9

Sample Output

Sum of Elements in Each Row:

Row 1: 6

Row 2: 15

Row 3: 24

Average of Elements in Each Column:

Column 1: 4.0

Column 2: 5.0

Column 3: 6.0

Standard Deviation of Elements in Each Row:

Row 1: 0.816497

Row 2: 0.816497

Row 3: 0.816497

III- DMA (Dynamic Memory Allocation): (15*2 = 30 Marks)

5. **Dynamic String Concatenation:** Build a C++ program that dynamically allocates memory for two strings and concatenates them into a new string using DMA. The program should allow users to input the two strings, and it should handle memory allocation efficiently.
6. **Student Records Database:** Design a program that manages student records (name, ID, and GPA) using dynamic memory allocation. Users should be able to add, delete, search for, and display student records. Ensure proper memory management.

Sample Input

1. Add Student
2. Delete Student
3. Search Student
4. Display Students
5. Exit

Enter your choice: 1

Enter student name: John

Enter student ID: 101

Enter student GPA: 3.75

Enter your choice: 1

Enter student name: Alice

Enter student ID: 102

Enter student GPA: 3.85

Enter your choice: 4

Student Records:

1. Name: John, ID: 101, GPA: 3.75

2. Name: Alice, ID: 102, GPA: 3.85

Enter your choice: 2

Enter student ID to delete: 101

Enter your choice: 4

Student Records:

1. Name: Alice, ID: 102, GPA: 3.85

Enter your choice: 5

IV- Strings (Array of characters): (10*2 = 20 Marks)

7. **Text Parsing:** Create a text parsing program in C++ that reads a sentence from the user and tokenizes it into words using arrays of characters. Implement a function to split the sentence into words and count the occurrences of each word.

Sample Input

Enter a sentence: This is a sample sentence. Is this a sample?

Sample Output

Word Frequencies:

This: 1

is: 2

a: 2

sample: 2

sentence.: 1

Is: 1

this: 1

sample?: 1

8. **String Encryption/Decryption:** Develop a program that encrypts and decrypts text using simple algorithms (e.g., Caesar cipher) with arrays of characters created using **DMA**. Users should be able to input a message, specify a key, and perform encryption or decryption.

Sample Input

Enter a message: Hello, World!

Enter a key (1-26): 3

Sample Output

Encrypted Message: Kloor, Zruog!